

FLAGSTAFF PULLIAM, AZ, USA

WMO: 723750

Lat: 35.144N

Lon: 111.666W

Elev: 7003

StdP: 11.34

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03103

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.4	9.9	-7.6	4.7	25.3	-3.7	5.8	26.1	28.3	33.1	25.5	34.0	1.2	40	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.1	85.9	55.1	83.6	54.8	81.4	54.6	61.5	72.5	60.3	72.1	59.4	71.6	10.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.4	95.0	63.8	57.0	90.3	63.1	55.6	85.7	62.6	30.8	72.2	29.9	71.9	29.1	71.3	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.7	22.0	19.0	-4.8	90.4	4.9	2.4	-8.3	92.2	-11.1	93.6	-13.8	94.9	-17.4	96.6		
(5)			-5.4	64.7	4.8	1.6	-8.8	65.9	-11.6	66.8	-14.3	67.7	-17.8	68.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	30.7	32.5	38.3	43.3	51.3	61.3	66.6	64.5	58.0	47.2	37.8	30.3		
	DBStd	14.17	7.06	7.24	6.39	6.38	6.42	5.85	3.57	3.34	4.68	6.03	6.68	7.41		
	HDD50	2832	599	491	362	211	62	3	0	0	4	122	366	612		
	HDD65	6744	1064	911	827	650	426	137	24	48	211	553	816	1077		
	CDD50	1699	0	0	0	11	102	341	514	451	246	34	0	0		
	CDD65	135	0	0	0	0	1	25	73	34	2	0	0	0		
	CDH74	3485	0	0	0	3	128	1125	1280	722	219	8	0	0		
	CDH80	775	0	0	0	0	15	276	340	128	16	0	0	0		
	WSAvg	6.6	6.3	7.0	7.3	8.6	8.1	7.7	5.4	4.5	5.3	6.1	6.4	6.5		
	PrecAvg	22.6	2.0	2.2	2.4	1.3	0.7	0.4	2.6	2.8	2.0	1.7	1.9	2.2		
(15) Precipitation	PrecMax	36.6	9.6	10.0	6.8	5.6	4.1	1.9	6.6	8.1	6.8	9.8	6.7	7.3		
	PrecMin	14.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0		
	PrecStd	6.1	1.9	2.0	1.9	1.2	0.8	0.5	1.4	1.6	1.8	1.8	1.7	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	61.6	67.5	72.9	82.2	89.0	89.5	86.4	82.1	75.0	67.0	58.4		
		MCWB	39.6	41.3	44.3	46.7	52.1	54.7	55.2	56.6	54.9	49.8	44.6	39.9		
	2%	DB	52.8	57.1	62.9	68.9	77.5	85.5	86.3	83.4	79.0	71.2	62.9	53.6		
		MCWB	37.3	39.4	42.7	45.3	49.8	53.4	56.0	56.7	53.9	48.5	43.0	38.2		
	5%	DB	49.4	52.7	59.4	65.3	74.1	83.3	83.8	80.9	76.5	68.3	59.1	49.5		
		MCWB	35.7	37.4	40.8	43.5	48.3	52.7	55.8	56.9	53.4	47.5	41.8	36.0		
	10%	DB	45.2	48.2	55.9	62.0	70.8	80.7	81.4	78.3	74.0	65.2	55.1	44.9		
		MCWB	34.2	35.7	39.4	42.2	46.9	51.8	55.8	57.0	53.3	46.3	40.1	34.3		
	0.4%	WB	41.9	43.6	45.9	48.6	53.5	58.8	63.2	63.2	61.1	54.6	47.2	42.7		
		MCDB	50.8	54.3	62.5	68.8	75.4	75.8	73.0	73.3	71.4	63.8	61.1	53.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	40.9	43.6	46.4	51.3	56.5	61.6	61.5	59.1	52.1	44.5	40.1		
		MCDB	48.6	52.8	60.9	65.8	72.5	77.5	72.6	72.6	69.6	64.1	57.8	49.4		
	5%	WB	37.0	38.9	41.7	44.4	49.8	55.0	60.4	60.4	57.5	50.0	42.8	37.5		
		MCDB	46.1	50.3	57.1	62.9	70.5	77.9	72.4	71.9	68.1	63.6	56.1	45.9		
	10%	WB	34.9	36.6	40.0	42.7	48.1	53.3	59.5	59.5	56.1	47.9	41.1	35.3		
		MCDB	43.7	46.6	54.0	60.2	68.2	76.5	72.2	71.4	67.1	62.1	53.1	43.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.1	24.2	27.2	28.7	31.2	34.4	28.1	26.5	29.1	29.8	27.4	24.3		
		MCDBR	31.8	33.9	35.6	37.3	37.6	37.3	33.3	31.4	34.0	37.5	36.3	32.7		
		MCWBR	18.9	19.2	18.6	18.6	17.7	16.5	13.0	12.3	14.7	19.1	20.3	19.5		
	5% WB	MCDBR	27.5	28.3	32.2	33.8	34.1	32.9	24.0	24.3	24.8	29.8	31.1	27.4		
		MCWBR	17.0	16.8	17.4	17.4	16.2	14.6	9.5	10.6	11.8	16.0	18.1	17.4		
(40) Clear-Sky Solar Irradiance	taub		0.232	0.235	0.254	0.264	0.278	0.280	0.326	0.328	0.283	0.268	0.238	0.232		
	taud		2.681	2.692	2.621	2.582	2.590	2.577	2.522	2.497	2.645	2.624	2.682	2.684		
	Ebn at Noon		320	329	327	325	319	316	301	299	312	310	315	314		
	Edh at Noon		22	24	28	31	31	31	33	33	27	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		1007	1306	1784	2226	2490	2651	2142	1980	1861	1492	1140	895		
	RadStd		89	111	116	100	160	129	137	105	71	111	56	63		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	-2.03	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)			+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+176	+72

Nomenclature: See separate page